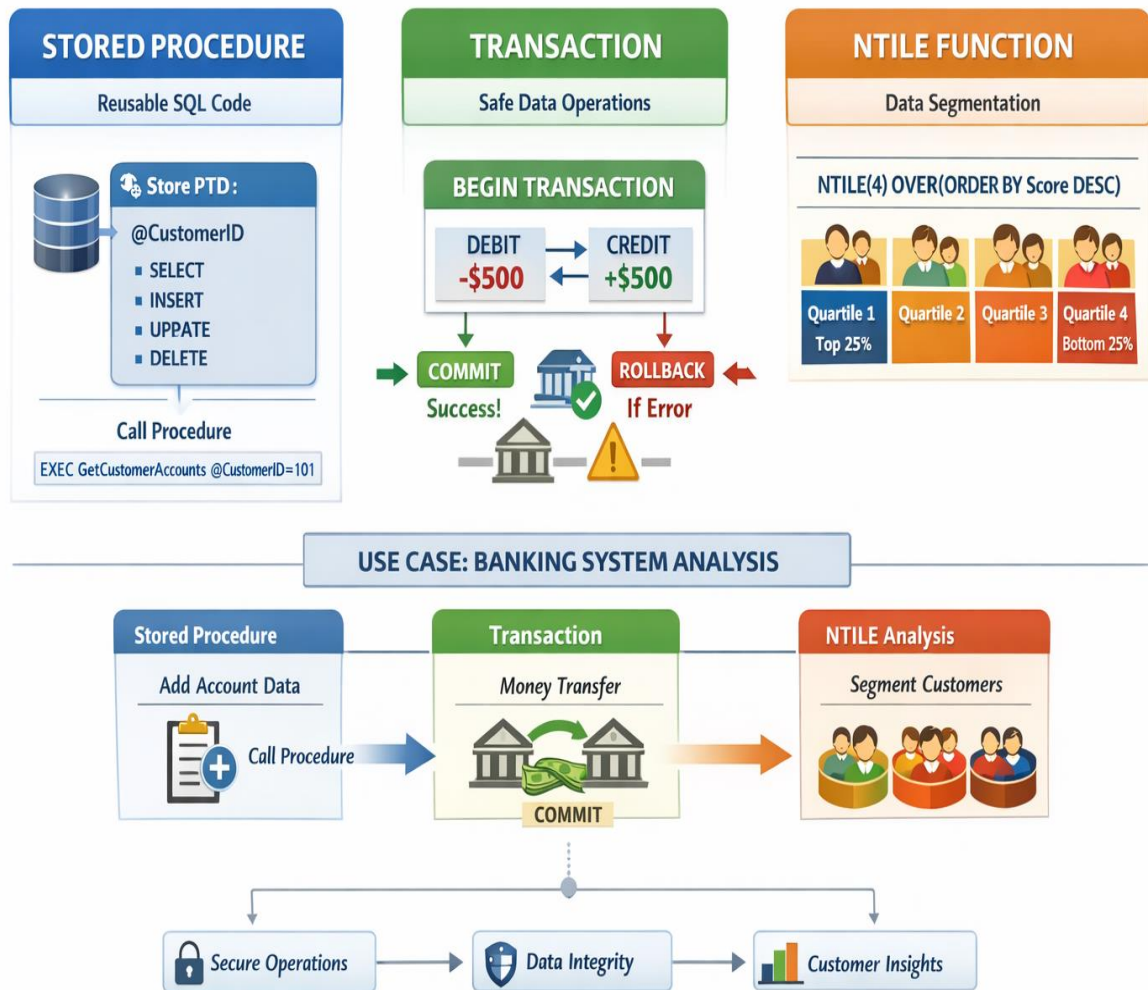


SQL Concepts Mini-Report

Stored Procedures - Transactions - NTILE Function



1. Stored Procedures

Definition:

A stored procedure is a set of SQL commands stored in the database that can be executed as needed.

Benefits:

- Simplifies repetitive tasks
- Improves performance (runs on the server)
- Enhances security (users execute procedure without direct table access)

Example in Diagram:

- Adds a new customer account using a procedure like:
`EXEC GetCustomerAccounts @CustomerID=101`

2. Transactions

Definition:

A transaction is a sequence of SQL operations executed as a single unit. All steps succeed (**COMMIT**) or all fail (**ROLLBACK**).

Benefits:

- Ensures safe and consistent data operations
- Prevents errors in sensitive processes, e.g., money transfers

Example in Diagram:

- Money transfer between accounts:
 - Success → **COMMIT**
 - Error → **ROLLBACK**

3. NTILE Function

Definition:

NTILE divides data into roughly equal groups (tiles) and assigns a group number to each row.

Benefits:

- Easy ranking and segmentation of data
- Useful for analytics and performance evaluation

Example in Diagram:

- Customers divided into 4 groups based on account balance:
 - Top 25% → Quartile 1
 - Bottom 25% → Quartile 4

Banking Use Case

- **Stored Procedure:** Quickly add or query customer accounts
- **Transaction:** Safely transfer money between accounts
- **NTILE:** Analyse and segment customers by balance

Outcome:

- **Secure Operations**
- **Data Integrity**
- **Customer Insights**